



DW02_mult_4_stage

Four-Stage Pipelined Multiplier

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- Parameterized word length
- Unsigned and signed (two's-complement) data operation
- Four-stage pipelined architecture
- Automatic pipeline retiming

Revision History

Description

DW02_mult_4_stage is a four-stage pipelined multiplier. DW02_mult_4_stage multiplies the operand A by B to produce a product (PRODUCT) with a latency of three clock (CLK) cycles.

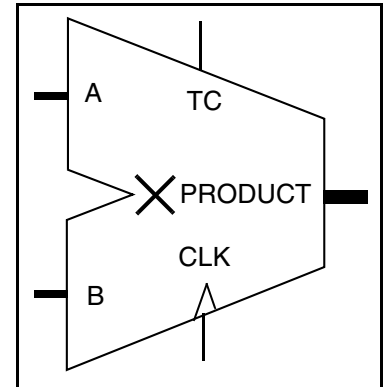


Table 1-1 Pin Description

Pin Name	Width	Direction	Function
A	A_width bits	Input	Multiplier
B	B_width bits	Input	Multiplicand
TC	1 bit	Input	Two's complement control ■ 0 = Unsigned ■ 1 = Signed
CLK	1 bit	Input	Clock
PRODUCT	$A_width + B_width$ bits	Output	Product ($A \times B$)

Table 1-2 Parameter Description

Parameter	Values	Description
A_width	≥ 1	Word length of A
B_width	≥ 1	Word length of B

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
str	Area or delay optimized flexible architecture	DesignWare

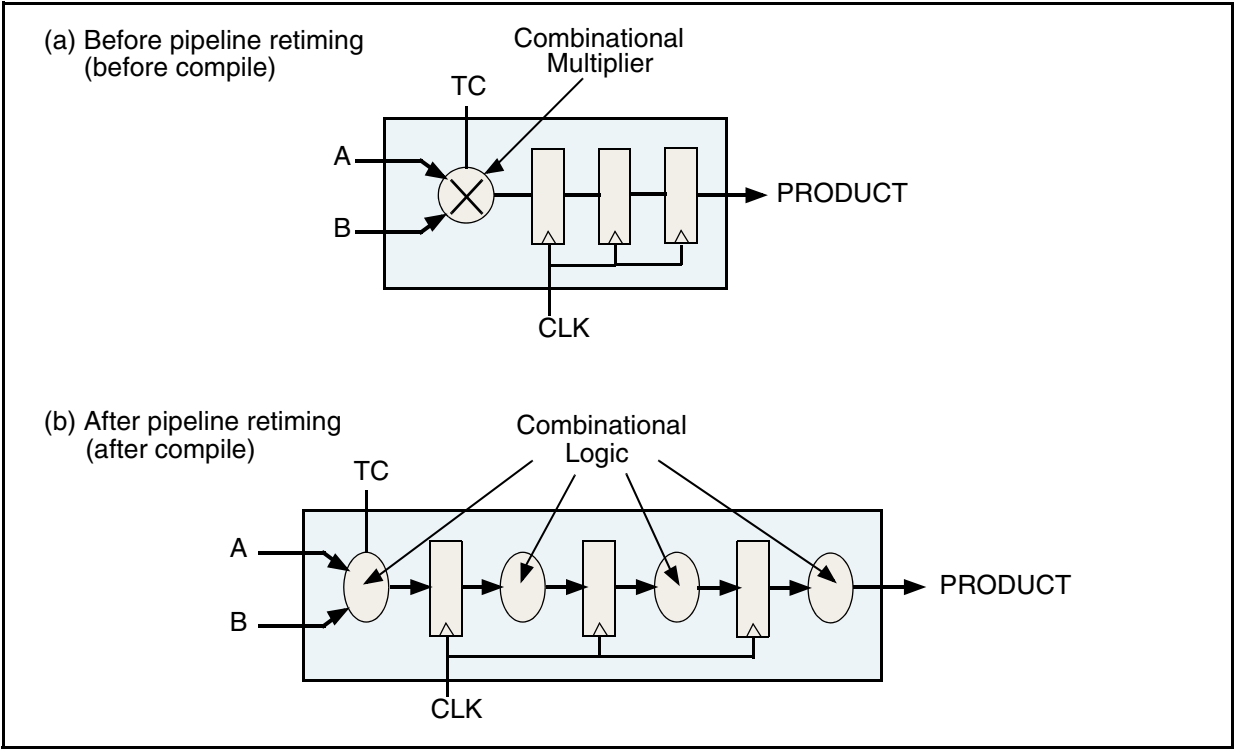
Table 1-4 Simulation Models

Model	Function
DW02.DW02_MULT_4_STAGE_CFG_SIM	Design unit name for VHDL simulation
dw/dw02/src/DW02_mult_4_stage_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW02_mult_4_stage.v	Verilog simulation model source code

The control signal, TC, determines whether the input and output data is interpreted as unsigned (TC = 0) or signed (TC = 1) numbers.

Automatic pipeline retiming ensures optimal placement of pipeline registers within the multiplier to achieve maximum throughput.

Figure 1-1 Block Diagram



Related Topics

- [Math – Arithmetic Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```

library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_comp.all;

entity DW02_mult_4_stage_inst is
  generic (inst_A_width : POSITIVE := 8;
           inst_B_width : POSITIVE := 8 );
  port (inst_A : in std_logic_vector(inst_A_width-1 downto 0);
        inst_B : in std_logic_vector(inst_B_width-1 downto 0);
        inst_TC : in std_logic;
        inst_CLK : in std_logic;
        PRODUCT_inst : out std_logic_vector(inst_A_width+inst_B_width-1 downto 0)
        );
end DW02_mult_4_stage_inst;

architecture inst of DW02_mult_4_stage_inst is
begin

  -- Instance of DW02_mult_4_stage
  U1 : DW02_mult_4_stage
    generic map ( A_width => inst_A_width, B_width => inst_B_width )
    port map ( A => inst_A,   B => inst_B,   TC => inst_TC,
              CLK => inst_CLK,   PRODUCT => PRODUCT_inst );
end inst;

-- pragma translate_off
configuration DW02_mult_4_stage_inst_cfg_inst of DW02_mult_4_stage_inst is
  for inst
  end for; -- inst
end DW02_mult_4_stage_inst_cfg_inst;
-- pragma translate_on

```

HDL Usage Through Component Instantiation - Verilog

```
module DW02_mult_4_stage_inst( inst_A, inst_B, inst_TC,
                               inst_CLK, PRODUCT_inst );

    parameter A_width = 8;
    parameter B_width = 8;

    input [A_width-1 : 0] inst_A;
    input [B_width-1 : 0] inst_B;
    input inst_TC;
    input inst_CLK;
    output [A_width+B_width-1 : 0] PRODUCT_inst;

    // Instance of DW02_mult_4_stage
    DW02_mult_4_stage #(A_width, B_width)
        U1 ( .A(inst_A),    .B(inst_B),    .TC(inst_TC),
            .CLK(inst_CLK),    .PRODUCT(PRODUCT_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
October 2019	DWBB_201903.5	■ Updated description of 'str' implementation in Table 1-3 on page 2
January 2019	DWBB_201806.5	■ Updated example in "HDL Usage Through Component Instantiation - VHDL" on page 3 ■ Added this Revision History table and the document links on this page

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